University of Michigan
Using Databases with Python

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Course Info

Using Databases with Python

by University of Michigan

About this Course

This course will introduce students to the basics of the Structured Query Language (SQL) as well as basic database design for storing data as part of a multi-step data gathering, analysis, and processing effort. The course will use SQLite3 as its database. We will also build web crawlers and multi-step data gathering and visualization processes. We will use the D3.js library to do basic data visualization. This course will cover Chapters 14-15 of the book “Python for Everybody”. To succeed in this course, you should be familiar with the material covered in Chapters 1-13 of the textbook and the first three courses in this specialization. This course covers Python 3.

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Taught by:

Charles Russell Severance
Clinical Professor, School of Information

Basic Info	Course 4 of 5 in the Python for Everybody Specialization
Level	Beginner
Commitment	5 weeks of study, 2-3 hours/week
Language	English Subtitles: Arabic, French, Bengali, Uzbek, Ukrainian, Chinese (Simplified), Greek, Italian, Portuguese (Brazil), Vietnamese, Dutch, Korean, German, Pashto, Urdu, Russian, Thai, Indonesian, Swedish, Turkish, Azerbaijani, Spanish, Dari, Hindi, Japanese, Kazakh, Hungarian, Polish
How To Pass	Pass all graded assignments to complete the course.
User Ratings	★ 4.8

Syllabus

Module 1

Chapter Fourteen: Object Oriented Python

To start this class out we cover the basics of Object Oriented Python. We won't be writing our own objects, but since many of the things we use like BeautifulSoup, strings, dictionaries, database connections all use Object Oriented (OO) patterns we should at least understand some of its patterns and terminology.

12 videos, 4 readings

1. Video: Welcome to Using Databases with Python
2. Reading: Help us learn more about you!
3. Reading: Python Textbook
4. Reading: Course Syllabus
5. Video: Unicode Characters and Strings
6. Reading: Notice for Auditing Learners: Assignment Submission
7. Video: Windows 10: Installing Python and Writing A Program
8. Video: Windows: Taking Screen Shots
9. Video: Macintosh: Using Python and Writing A Program
10. Video: Macintosh: Taking Screen Shots
11. Video: Object Oriented Definitions and Terminology
12. Video: Our First Class and Object
13. Video: Object Life Cycle
14. Video: Object Inheritance
15. Video: Bonus: Interview - Software Engineering - Bertrand Meyer
16. Video: Bonus: Office Hours - London

Graded: Using Encoded Data in Python 3
Graded: Object Oriented Programming

View Less

Module 2

Chapter Fifteen: Basic Structured Query Language

We learn the four core CRUD operations (Create, Read, Update, and Delete) to manage data stored in a database.

6 videos

1. Video: Relational Databases
2. Video: Using Databases
3. Video: Single Table CRUD
4. Video: Worked Example: Counting Email in a Database
5. Video: Bonus: Office Hours Zagreb, Croatia
6. Video: Interview: Elizabeth Fong - The Early Years of SQL

Graded: Single-Table SQL
Graded: Our First Database
Graded: Counting Email in a Database

Module 3

Chapter Fifteen: Data Models and Relational SQL

In this section we learn about how data is stored across multiple tables in a database and how rows are linked (i.e., we establish relationships) in the database.

8 videos

1. Video: Designing a Data Model
2. Video: Representing a Data Model in Tables
3. Video: Inserting Relational Data
4. Video: Reconstructing Data with JOIN
5. Video: Worked Example: Tracks.py
6. Video: Bonus: Office Hours Perth, Australia
7. Video: Bonus Interview: Niklaus Wirth
8. Video: Bonus: Office Hours Barcelona

Graded: Multi-Table Relational SQL
Graded: Multi-Table Database - Tracks

Module 4

Chapter Fifteen: Many-to-Many Relationships in SQL

In this section we explore how to model situations like students enrolling in courses where each course has many students and each student is enrolled in many courses.

4 videos

1. Video: Many-to-Many Relationships
2. Video: Worked Example: roster.py
3. Video: Bonus: Office Hours Mexico, City
4. Video: Bonus Interview: Andrew Tannenbaum - Minix

Graded: Many-to-Many Relationships and Python
Graded: Many Students in Many Courses

Module 5

Chapter Sixteen: Databases and Visualization

In this section, we put it all together, retrieve and process some data and then use the OpenStreetMaps API to visualize our data.

5 videos, 3 readings

1. Video: Geocoding
2. Video: Worked Example: Geodata
3. Reading: Peer Graded Assignment - Instructor Input
4. Video: Bonus: Office Hours - Amsterdam
5. Video: Bonus Interview: Richard Stallman - Free Software Foundation
6. Video: Bonus Interview: Brian Behlendorf - Apache Foundation
7. Reading: Please Rate this Course on Class-Central
8. Reading: Post-Course Survey

Graded: Databases and Visualization (peer-graded)

How It Works

General

What do start dates and end dates mean?
Once you enroll, you’ll have access to all videos, readings, quizzes, and programming assignments (if applicable). If you choose to explore the content without purchasing, you may not be able to access certain assignments. If you don’t finish all graded assignments before the end of the session, you can reset your deadlines. Your progress will be saved and you’ll be able to pick up where you left off.

What are due dates? Is there a penalty for submitting my work after a due date?
Within a course, there are suggested due dates to help you manage your schedule and keep work from piling up. Quizzes and programming assignments can be submitted late without consequence. However, it is possible that you won't receive a grade if you submit your peer-graded assignment too late because classmates usually review assignment within three days of the assignment deadline.

Can I re-attempt an assignment?
Yes. If you want to improve your grade, you can always try again. If you’re re-attempting a peer-graded assignment, re-submit your work as soon as you can to make sure there’s enough time for your classmates to review your work. In some cases you may need to wait before re-submitting a programming assignment or quiz. We encourage you to review learning material during this delay.

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Course 4 of Specialization

Learn to Program and Analyze Data with Python

Develop programs to gather, clean, analyze, and visualize data.



University of Michigan
Python for Everybody

View the course in catalog

Related Courses



University of Michigan
Python Data Structures



University of Michigan
Using Python to Access Web Data



University of Michigan
Programming for Everybody (Getting Started with Python)